**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Validation and Authentication

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Which integers can random(1,6) produce in OCR ERL?

2. What is HTTP used for?

A) Sending outgoing email
B) Syncing an email inbox
C) Transferring web resources
D) Looking up IP addresses

3. "Penetration testing permits anyone to attack any network without permission." Correct this. Explain why.

4. A zero-indexed 2D array is [[11,12],[21,22]]. Give grid[1,0], then state the indexes needed to access 12.

## Today's objectives

Apply validation and authentication to programs. Apply this to another software scenario.

## Key vocabulary - Read, cover, recall

**Validation**

Checks input against specified rules, rejecting values outside the allowed range.

**Range check**

Tests whether a numeric value is between minimum and maximum limits.

**Authentication**

Checks identity, for example using username and password, rather than checking input format.

**Re-prompt**

A valid design asks again when input does not meet a validation rule.

**Write one key term and its meaning in your own words:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Validation** | Checks input against specified rules, rejecting values outside the allowed range. Consider this in a online age entry. |
| **Range check** | Tests whether a numeric value is between minimum and maximum limits. Consider this in a exam score entry. |
| **Authentication** | Checks identity, for example using username and password, rather than checking input format. Consider this in a club registration. |
| **Re-prompt** | A valid design asks again when input does not meet a validation rule. Consider this in a account login. |

## I Do - One worked example

**A online age entry needs validation and range check. Explain why. [4 marks]**

1. Identify the inputs, constraints and expected result.

2. Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1].

3. Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1].

4. Give a specific point and a linked reason for each feature.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a exam score entry, explain the purpose of validation. [2 marks]

2. A club registration needs range check. Explain its role. [2 marks]

3. A account login uses authentication and re-prompt. Explain both. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Complete the validation expression so an octet of 256 is rejected.**

```
octet = 256
if ... then
print("Accepted")
else
print("Rejected")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how range check and authentication help in a school trip form. [4 marks]

Q4. Explain the roles of validation, authentication and re-prompt in a café membership. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Validation checks inputs against defined rules and can reject values outside the allowed range. ☐ T ☐ F

2. Checks input against specified rules, rejecting values outside the allowed range ☐ T ☐ F

3. Tests whether a numeric value is between minimum and maximum limits ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Any integer from 1 through 6 inclusive. ☐

**Q2** C - Transferring web resources ☐

**Q3** Testing with permission, within agreed limits, to find and fix security weaknesses. ☐

**Q4** (a) grid[1,0] is 21; 12 is at grid[0,1]. ☐

## Code First - Model

```
octet = 256
if octet >= 0 AND octet <= 255 then
print("Accepted")
else
print("Rejected")
endif
```

**Output: Rejected**

## We Do - Check the model

1. Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1]. ☐

2. Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1]. ☐

3. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. Re-prompt [1]; A valid design asks again when input does not meet a validation rule. [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Validation [1]; Authentication [1]. ☐

**Q3** Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1]. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. ☐

**Q4** Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1]. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. Re-prompt [1]; A valid design asks again when input does not meet a validation rule. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**